Palaeomagnetism of the Great Whin Sill

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Summary

The Great Whin Sill, which was intruded in Upper Carboniferous to Lower Permian times, has been sampled at 36 sites in Northumberland and Durham. It was found to be magnetically stable, the site mean direction of magnetization being 187°·8–4°·9 and the mean pole position lying in latitude 37°·3 N longitude 168°·9 E. The distribution of the mean directions of magnetization at these sites is not circular but oval, with the variation in the direction of the palaeomagnetic longitude about twice that in the direction of palaeomagnetic latitudes. However the poles calculated from the site mean directions of magnetization have a circular distribution, and this observation leads to proposal of a method of calculating the errors associated with the palaeomagnetic pole positions.

1. Introduction

The objects of this investigation were to determine the mean direction of magnetization of the Great Whin Sill and to investigate the pattern of secular variation. It has been argued (Watson & Irving 1957) that random errors are averaged out in the calculation of site-mean directions of magnetization, the latter representing spot readings of the geomagnetic field in an igneous body such as the Whin Sill. The distribution of the site mean directions of magnetization about their mean should therefore give the pattern of secular variation at the latitude in which the formation cooled through its Curie temperatures. As the Whin Sill is a composite intrusion and fairly thick, it seems probable that its cooling history would cover a time range sufficiently long such that its remanent magnetization would record the full range of secular variation.

In the statistical analysis of most previous work in palaeomagnetism it has been implicitly assumed that the directions of magnetization within a formation are circularly distributed around the mean direction, as proposed by Fisher (1953). Cones of confidence of half angle α within which the true mean directions of magnetization lie at given probabilities have been derived. The mean pole positions for the times represented by these formations have been computed from the mean directions of magnetization and areas of confidence have been constructed around them (Creer & others 1957). For formations magnetized at the poles these areas are circular in shape and of radius 2α. At the equator, they are ellipses with axes of half-length α and $\frac{1}{2}\alpha$, the latter in the direction of co-latitude. Between

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these two extremes of latitude their form is intermediate. Thus the shape of the areas of confidence so calculated depends on the palaeo-latitude. However, if we assume a circular distribution of the pole positions calculated for each site within a formation about the mean pole position for the formation as a whole, the observed directions of magnetization should have an oval distribution, this becoming more marked at low latitudes. Intense sampling of the Whin Sill was carried out to determine whether the mean directions of magnetization of sites or the mean pole positions of sites have circular distributions. This formation was chosen because earlier work by Hallimond & Butler (1949), and Manley (1949), had shown it to be almost horizontally magnetized. Furthermore the geology is well known and many natural and artificial outcrops are easily accessible.

2. The geology of the Whin Sill

The Great Whin Sill is the name given to a system of transgressive quartz dolerite sheets intruded into the Carboniferous rocks of northern England between latitudes 54°·6 N and 55°·6 N, and longitudes 1°·6 W and 2°·5 W (Figure 1). The outcrops sweep across Northumberland southward into County Durham, forming crags which, near Haltwhistle, carry the Roman Wall. It is also exposed in the west-facing Pennine escarpment. Associated with the Whin Sill is a system of petrologically similar dykes which may in part have been feeder dykes. The average thickness of the Whin Sill where it occurs as a single sheet is from 80 to 100 ft (Woolacott 1923), but as variation is both rapid and proportionately great, average thickness has little meaning.

The age of the intrusive system can be found within fairly close limits. The sill is generally intruded into lower Carboniferous rocks but associated dykes have been found cutting the Middle Coal Measures (Westphalian). In north Northumberland, at least, intrusion post-dates the folding of the lower Carboniferous beds (Westoll & others 1955). The upper limit, suggested by the fact that no dykes are known to cut Permian rocks, is better established by the discovery of rare pebbles of Whin type in the Upper Brockram of the Eden Valley. It can also be shown that intrusion preceded the mineralization of northern England (Holmes & Harwood 1928) and since detrital heavy mineral grains from the latter source have been found in the Permian Yellow Sands, the upper age limit can be regarded as reasonably fixed. In zonal terms the intrusion must be post-Westphalian pre-Thuringian, that is either Stephanian, Autunian or Saxonion, and Holmes and Harwood conclude "The interjection of the Whin Sill and related dykes may thus be regarded as the closing act of the Upper Carboniferous in the north of England". As will appear later in this paper, there are reasons for suggesting that an early Permian age is equally likely.

The Whin is a hard blue rock very much in demand as a road metal. Petrologically, the dominant Whin type is a medium-grained quartz dolerite. The occurrence of a coarse-grained dolerite pegmatite found as flattened lenticular masses in the upper parts of thick outcrops appears to be restricted to the south of the Tyne (Tomkeieff 1928). At contacts, a fine-grained marginal selvage, or occasionally vesicular, or even "white" Whin may be developed. Contacts of Whin against Whin, evidence of multiple intrusion, are known. Whin inclusions in Whin have also been described by Smythe (1914) as further evidence of multiple injection.

The average grain size of the normal type ranges from 0·2 to 1 mm. Microscopically, the structure is that of an interlacing network of plagioclase laths and

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elongate or stumpy prisms of augite held together by plates or granular aggregates of augite with iron ore minerals in skeletal or well-individualized form. Ophitic, intergranular and intersertal textures are all well developed, frequently within a single section. Minor amounts of alkali felspar, quartz and micro-pegmatite occur interstitially with hornblende, biotite and accessory apatite. Chloritic and serpentinous minerals are the common alteration products. The sills are characterized by their homogeneity, the range of variation is so small that the diversity at a single exposure is comparable to that of the whole extent of the sill (Tomkeieff 1928).

From the magnetic point of view the important factors are the amount and composition of the opaque ores for which, on the whole, information is less complete. Micrometric measurements indicate that the opaque ores range from 5 to 10 per cent of the rock by volume, although according to Holmes & Harwood (1928) these estimates may be in error by as much as 50 per cent, particularly in the finer grained varieties. The normative magnetite calculated from chemical analyses averages from 4 to 5 per cent, while normative ilmenite is about 4 per cent. Akimoto (1955) has studied the ferromagnetic minerals in a typical specimen of Whin Sill and gives the cell dimension $a = 8{\cdot}415$Å (pure magnetite $a = 8{\cdot}413$Å) and a single Curie point at 565 °C (pure magnetite 575 °C) from which it appears that the magnetic mineral is very close to pure magnetite. If ilmenite occurs it must be as a separate paramagnetic phase. Pyrrohitite has only been recorded around inclusions of Whin materials within a Whin dyke (Smythe 1914).

The Whin intrusions have produced thermal metamorphic minerals in narrow zones at the contacts. Recrystallization occurs in some limestones while sandstones may locally become quartzites, and hornfelses may even develop. Garnet, wollastonite, idocrase and andalusite have all been found.

3. Method of sampling and measurement

Samples of the dolerite have been taken at 36 sites (quarries, sea cliffs or inland scarps) distributed along the outcrop (Figure 1). Three samples were obtained from each site, great care being taken to select fresh material. An exposed face of each was oriented by standard geological methods before extraction. A single core $1\frac{3}{8}$ in. in diameter and height with its axis perpendicular to the orientated face was machined from each sample using non-magnetic tools. The directions of magnetization of the cores were measured by a short period astatic magnetometer of the type described by Collinson & others (1957). The overall accuracy of orientation and measurement is 5°. Errors from this source are random, since the orientated surfaces were selected at random, and are therefore eliminated when the average direction of several specimens is taken. Of the 108 samples collected, three from a single site (No. 34) gave random directions and were rejected, while three others distributed among three other sites broke up during machining and were not used.

At three sites (16, 17 and 19), several samples of sedimentary rocks were collected from near the lower contact. From each of these samples a single one-inch core was cut and then sliced into several disks. The directions of magnetization of the disks were measured by a sensitive astatic magnetometer of the type described by Collinson & others (1957).

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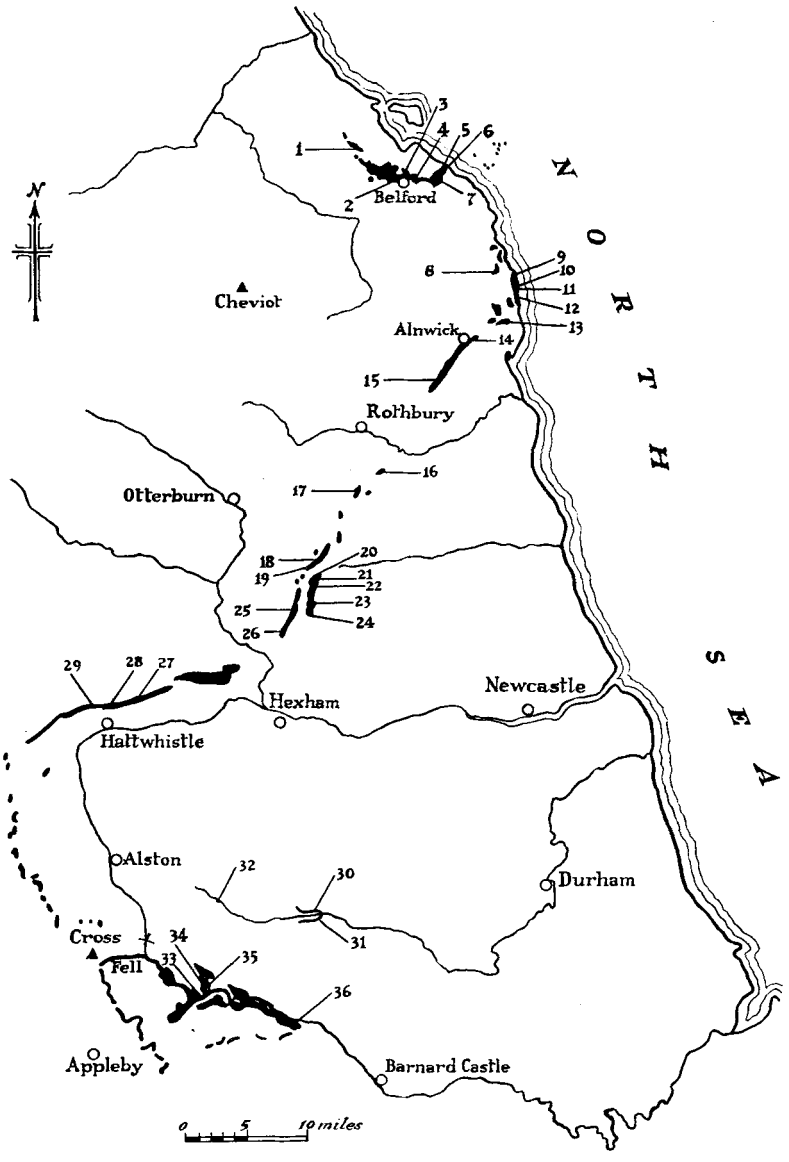


FIG. 1.—Map showing the distribution of sampling sites along the outcrop of the Whin Sill of north-eastern England.

4. The magnetization of the dolerite

At any one place the sills are either horizontal or dip at low angles, generally less than 10°. As it is uncertain whether tilting everywhere preceded or post-dated injection, the directions of magnetization have been worked out in two ways:

(a) with the beds in their present position, the assumption being that injection occurred after folding;

(b) with the sills everywhere reduced to a horizontal plane on the assumption that tilting occurred after intrusion.

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The directions of magnetization in the 102 cores of dolerite are given according to the two methods in Figures 2a and 2b respectively. The plots are on stereographic equatorial projections so that positive and negative dips do not obscure one another and the shape of the distribution can be more clearly seen. The directions are consistently southerly with low dips and make angles of over 90° with the direction of the present field. Statistical tests as described by Watson & Irving (1957) show that the two distributions (a) and (b) are not Fisherian. The dispersion appears to decrease very slightly after correction for dip. The mean site directions, corrected to a horizontal plane, are plotted in Figure 3 with the overall

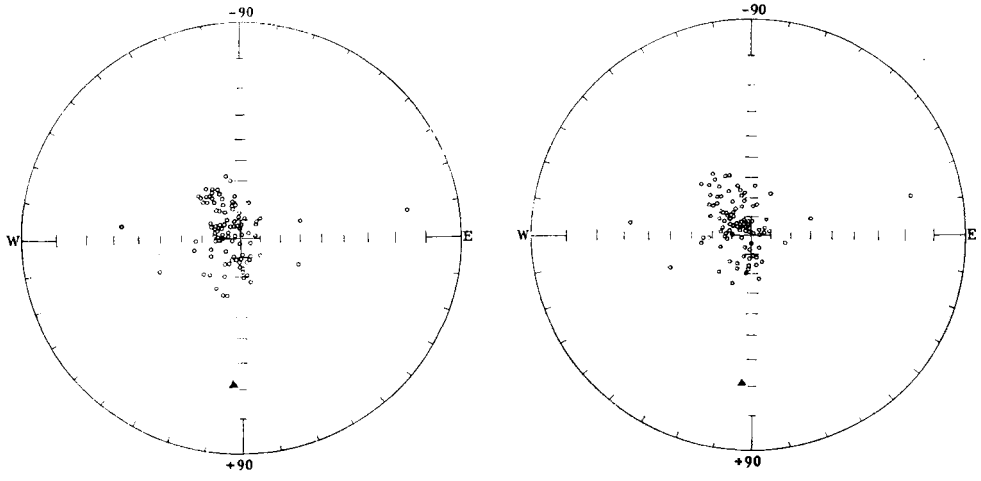


FIG. 2.—Stereographic equatorial plots of the directions of magnetization of Whin Sill samples.
(a) Results plotted with respect to the present field position.
(b) Results plotted with the Whin Sill restored to a horizontal plane.

mean direction at the centre of the stereogram. For both site and specimen directions the distribution of individual values around the mean direction is oval not circular with the variation in inclination about twice that in declination.

Certain magnetic properties of samples from five sites chosen at random were measured. Samples were placed in a d.c. magnetic field and the remanent magnetization (i.r.m.) on removal from the field was measured. The strength of the d.c. field was increased in steps and it was found possible to saturate the samples. The maximum i.r.m.s so obtained for samples from the five sites are recorded in Table 2. Next the samples were placed in d.c. magnetic fields directed in opposition to the maximum i.r.m. so obtained. The reduced intensity of remanent magnetization was measured for successively higher d.c. field strengths. The field strength required to remove the maximum i.r.m. is here defined as the coercivity of remanence and is also recorded in Table 2. The values of the two above quantities are similar to those previously given for basic igneous rocks (Nagata 1953, p. 107 and Creer 1955) and indicate that the magnetic minerals present are predominantly of the magnetite rather than the haematite family.

The initial susceptibilities (in d.c. fields of the order of the Earth's field) of samples from the same five sites were also measured. The values, also recorded

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Table 1

Directions of magnetization of the Whin Sill

Site No.	Locality	National Grid reference	Site Directions of Magnetization (a) referred to present horizontal		(b) corrected to bedding		Site pole position (from b)	
			D	*I*	*D*	*I*	Lat. (°N)	Long. (°E)
1	Kyloe Hills, Belford, natural cliff outcrop	NU 042 395	177	+ 2	176	−10	39°	183°
2	Belford, natural cliff near Craggyhall	NU 103 346	185	− 2	185	− 6	37°	172°
3	Belford, quarry	NU 110 346	195	− 5	196	− 5	35°	158°
4	Belford Station quarry	NU 129 342	178	−11	179	− 4	37°	180°
5	Bamburgh, sea cliffs, near lighthouse	NU 173 360	187	− 7	186	− 5	37°	171°
6	Bamburgh, sea cliffs, under castle	NU 184 352	192	+ 7	192	+ 3	32°	164°
7	Budle quarry, near Bamburgh	NU 160 342	183	− 2	183	− 1	35°	175°
8	Embleton quarry	NU 232 230	203	−10	203	−11	36°	150°
9	Dustanburgh castle, sea cliffs	NU 260 221	191	−22	193	−28	48°	159°
10	Craster harbour, sea cliffs	NU 259 200	169	−16	171	−14	38°	190°
11	Craster quarry, sea cliffs	NU 257 197	190	−17	191	−14	40°	164°
12	Cullernose Point, sea cliffs	NU 260 187	189	−25	190	−20	44°	165°
13	Longhoughton, Howlet Hills quarry	NU 234 956	187	−15	187	−15	42°	169°
14	Longhoughton, Snableazes quarry	NU 224 139	199	−22	203	−26	46°	153°
15	Newton on the Moor, Hampeth burn quarry	NU 154 075	196	−19	196	−19	44°	170°
16	Rothbury, Ward's Hill quarry	NU 078 964	205	+ 6	205	+ 6	28°	150°
17	Rothbury, Coltpark quarry	NZ 061 941	182	+ 7	182	+ 9	30°	176°
18	Knowesgate quarry	NY 991 858	189	− 5	190	−13	41°	165°
19	Knowesgate, West Whelpington Crag's quarry	NY 977 838	183	−10	183	+ 1	34°	174°
20	Kirkwhelpington river cliffs	NY 999 843	193	+28	190	+15	37°	167°
21	Kirkwhelpington, Three Farms quarry	NY 988 834	184	+12	183	+ 7	31°	174°
22	Kirkwhelpington, Northside natural outcrops	NY 986 824	180	+15	179	+11	29°	179°
23	Great Bavington, natural cliff outcrop	NY 982 807	189	+14	190	+13	38°	167°
24	Little Bavington, Divethill quarry	NY 980 790	183	− 9	177	−10	40°	182°
25	Thockrington, natural cliff outcrop	NY 957 790	184	−17	195	+11	28°	161°

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Table 1—*continued*

Site No.	Locality	National Grid reference	Site Directions of Magnetization (a) referred to present horizontal		(b) corrected to bedding		Site pole position (from b)	
			D	*I*	*D*	*I*	Lat. (°N)	Long. (°E)
26	Gunnerton quarry, near Barrasford	NY 912 744	182	+12	182	+11	29°	176°
27	Twice Brewed, natural cliff outcrop	NY 754 676	182	+ 7	182	+ 7	31°	175°
28	Greenhead, quarry near Milecastle 42	NY 715 667	192	+ 9	190	−15	42°	164°
29	Greenhead, Walltown quarry	NY 671 661	183	−15	183	−20	45°	173°
30	Stanhope, Greenfoot quarry	NY 982 391	194	−11	194	− 7	38°	160°
31	Stanhope, Tees river bed	NY 988 391	186	−10	186	− 6	38°	170°
32	Copthill, quarry (laccolith)	NY 836 419	202	+18	202	+18	—	—
33	High Force, Tees river cliff	NY 872 281	192	− 7	192	− 9	38°	162°
34	Langdon Beck quarry 1	NY 871 291	—	—	—	—	—	—
35	Langdon Beck quarry 2	NY 875 292	192	+ 3	192	0	35°	163°
36	Middleton in Teesdale, Crossthwaite quarry	NY 836 418	181	+11	180	0	35°	178°
	Mean values ($N = 34$)				187°·8	−4°·9	37°·3	168°·9

Table 2

Site	Maximum i.r.m. gauss ($\times 10^{-1}$)	Field required to saturate (oersteds)	Coercivity of remanence (oersteds)	n.r.m. gauss ($\times 10^{-4}$)	Susceptibility in $H = 0{\cdot}49$ oe e.m.u. ($\times 10^{-4}$)	Ratio of n.r.m. to susceptibility
7	6·9	1 000	270	19·9	22·5	0·9
10	13·1	1 000	350	31·0	21·6	1·4
12	18·9	1 800	390	40·0	23·1	1·7
15	10·0	1 300	330	28·6	16·4	1·7
24	7·5	800	270	20·2	15·6	1·3

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in Table 2 are similar to those measured by Irving (1956) in the Tasmanian dolerite sills which are of similar composition. The same is true of the intensities of n.r.m.

5. Contact sediments and origin of magnetization of the sills

At three sites (16, 17 and 19) samples of sedimentary rocks near the lower Whin contact have been collected. The upper contacts when exposed are generally inaccessible. At site 19, a continuous section was obtained through the exposed thickness of two feet of fine-grained yellow-brown sandstone. At site 17, a greater

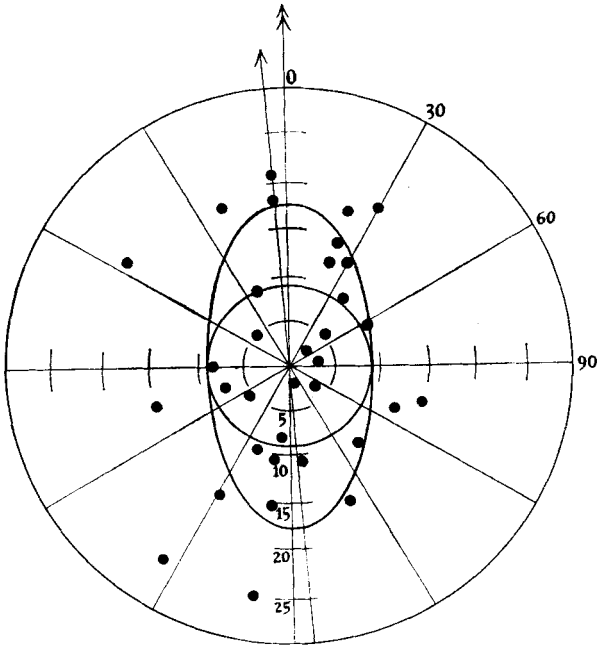


FIG. 3.—Centred stereographic plots.
Site mean directions of magnetization. The semi-axis of the oval (9°·7 and 19°·3) have been calculated from the standard deviation of the distribution of poles as described in paragraph 8. 21 points lie inside this oval and 13 outside.

thickness of more massive ganister-like sandstone is exposed and samples were taken at the contact and 18 in. lower down. Similarly at site 16 two samples of bluish limestone were collected at, and 18 in. below, the contact. The angular differences between the mean directions of the contact sediments and the Whin Sill at each of the three sites were 4, 14 and 16° respectively. Close to the contact the intensity of n.r.m. is about 10^{-5} gauss and 2 ft away is about 10^{-6} which is an order greater than in unheated sediments of the same type. The latter are so weakly magnetized as to be only just measurable. The higher n.r.m. parallel to that of the sills found in the contact sediments suggests that both magnetizations are thermo-remanent in origin and that they consequently date from the time of injection and cooling of the latter.

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6. Stability of n.r.m. of the Whin Sill

Ten specimens were randomly selected and the directions of n.r.m. were measured again after an interval of one year, having been stored in the Earth's field during that time. The differences between original and repeat directions of magnetization of the specimens were 1°, 13°, 4°, 15°, 15°, 7°, 26°, 7°, 4°, and 0°. Although these may seem unduly great, the difference of 7° between the original and repeat mean directions of this group of specimens is not significant as the standard errors are 6°·2 and 5°·3 respectively. These differences are probably due to a randomly directed temporary magnetization acquired during storage in the laboratory.

Ten different specimens from the same sites as the remeasured samples were subjected to a.c. demagnetization as described by Creer (1958 and 1959). Demagnetization curves for two of these specimens are shown in Figure 5 (curves 1 and 2) the other eight being cleaned by an alternating field of 280 oersteds peak value without treatment in fields of intermediate strength. Specimens showed differences in direction of magnetization before and after cleaning of 9°, 7°, 5°, 9°, 23°, 35°, 6°, 6°, 5° and 17°. These changes in direction are irregular so that the difference between the mean direction of magnetization of these specimens before and after cleaning is only 3° and is not significant.

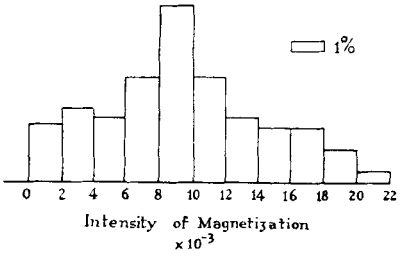


FIG. 4.—Histogram of the intensity of magnetization of Whin Sill samples.

The specimens showing the biggest changes in each of the above tests came from site 32, and because of doubt as regards the stability of this rock, no pole position has been calculated from the site mean direction of magnetization, nor has the latter been used in the statistical computation of the mean direction of magnetization of the whole formation.

In Figure 5, curve 3 illustrates the behaviour of a highly unstable rock (a Triassic basalt from Uruguay studied by Creer 1958), a marked decrease in intensity and a marked change in direction of magnetization being produced by demagnetizing fields of peak value less than 100 oersteds. The Whin Sill specimens did not behave in this way and the magnetization of this formation is therefore believed to be stable within the standard deviation of the distribution about the mean.

7. Length of record

In a single sill all parts do not cool through the Curie point at the same time. Jaeger (1957) has shown, on reasonable assumptions, that after injection the margins cool to about 665 °–620 °C which is somewhat higher than the Curie temperature of magnetite (580 °C). Thereafter the temperature falls slowly, the time taken

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for the Curie point isotherm of magnetite to pass from the margin to the centre being approximately $0{\cdot}006d^2$ years where d is the thickness in metres. For the Whin Sills this is of the order of 10 years. Thus the average of direction from several specimens taken from one locality—say for instance a quarry—will approximate to a spot reading of the field direction. However it is unlikely that the sills were injected at the same time, and in some places there is evidence of multiple intrusion (Westoll & others 1955, Smythe 1914). The time spread is probably not less than a hundred years and is more likely to be several thousand years. Therefore the directions of magnetization at different sites are likely to be affected by the secular variation of the field.

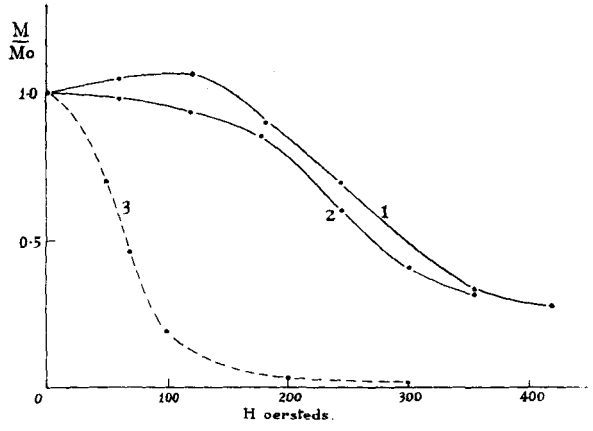


FIG. 5.—Demagnetization curves of Whin Sill (curves 1 and 2) and Uruguay basalt (curve 3) samples.

8. Analysis of directions of magnetization

It has been customary to use the distribution suggested by Fisher (1953) to compute the mean direction and confidence limits for palaeomagnetic directions obtained from a rock formation. From this it is an easy matter to calculate the attitude of the mean geocentric dipole (relative to the present geographical coordinates) in the field of which the rock formation became magnetized (Creer, Irving & Runcorn 1957). Fisher's distribution supposes that the individual directions, when regarded as points on a unit sphere, are distributed over the sphere with probability density

$$K \exp(K \cos \theta)/4\pi \sinh K \qquad (1)$$

where θ is the angle between the mean direction (direction cosines λ, μ, ν) and the direction of a specimen (l, m, n) and is given by

$$\cos \theta = \lambda l + \mu m + \nu n \qquad (2)$$

the individual directions are symmetrically disposed around the mean direction so that the contours containing given percentages of the observations are concentric circles with their centres at the true mean direction. Their radius θ_0 in radians is given by

$$\theta_0 = \{-2/K . \ln[\text{Prob}(\theta > \theta_0]\}^{\frac{1}{2}} \qquad (3)$$

provided θ_0 is fairly small.

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An estimate k of K is given by

$$k = N-1/N-R \tag{4}$$

where

$$R^2 = (\Sigma l_i)^2+(\Sigma m_i)^2+(\Sigma n_i)^2. \tag{5}$$

For a Fisherian distribution the angle in radians equal to $(2/K)^{\frac{1}{2}}$ is analagous to the standard deviation σ in a Gaussian distribution and in the following will be termed standard deviation and denoted by θ_0. Thus

$$\theta_0 = 81/K^{\frac{1}{2}} \text{ degree} \tag{6}$$

as has previously been derived by Runcorn (1957) and, for N sufficiently large that $N \doteqdot N-1$, is the same as the analogue of standard deviation used by Wilson (1959). However, whilst for a Gaussian distribution about 68 per cent of the measurements lie within the standard deviation, equation (3) shows that only about 63 per cent of the measurements lie within the analogous quantity for a Fisherian distribution.

It can be seen, by inspection of Figures 2 and 3, that this distribution is not applicable to the directions observed in the Whin Sill which have an oval and not a circular distribution.

The directions of the Earth's palaeomagnetic field measured for a number of sites do not give a continuous record of the field but a series of spot readings of the field during the time the rock formation in question became magnetized. Thus, as an alternative we may represent the directions at one site by a geocentric dipole (f), and assume that this dipole undergoes secular oscillations in a random fashion around the mean dipole (F). Thus in equation (1) θ becomes the angle between f and F, and the distribution of observed directions of magnetization will be oval, the semi-axes depending on the palaeomagnetic latitude of the rock formation as follows:

$$\delta D = \theta_0(1+3\cos^2\psi)^{-\frac{1}{2}} \tag{7}$$

and

$$\delta I = 2\theta_0(1+3\cos^2\psi)^{-1} \tag{8}$$

where ψ is the angular distance from the rock site to F (palaeomagnetic colatitude) and δD and δI are the changes in palaeomagnetic declination and inclination caused by movement of the pole from F to f. In low palaeomagnetic latitudes, δD tends to $81K^{-\frac{1}{2}}$ and δI tends to $162K^{-\frac{1}{2}}$ for $P(\theta < \theta_0) = 0{\cdot}63$. In high geomagnetic latitudes both δD and δI tend to $40{\cdot}5\ K^{-\frac{1}{2}}$. Thus, given a circular distribution of poles (f), the observed directions of magnetization of rocks from low palaeomagnetic latitudes should have oval distributions with $\delta D = \frac{1}{2}\delta I$ while in rocks from high palaeomagnetic latitudes the observed directions of magnetization should have a circular distribution, the variation in inclination and declination being one-half the variation for declination in low latitudes.

The inclination of the mean direction of magnetization in the Tasmanian dolerites is 85° indicating formation at high latitudes, and Watson & Irving (1956) found that the site directions of magnetization were distributed about the mean in a Fisherian fashion. The results presented in this paper for the Whin Sill provide data for low palaeomagnetic latitudes. The distribution of site directions of magnetization can be seen to be oval (Figure 3) and not circular. However, poles (f) for each of the 34 sites have been calculated and are plotted on a

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stereogram with the mean pole F at the centre of the projection. The distribution is seen to be more nearly circular (Figure 6) which suggests that it is more correct to apply Fisherian statistical methods to site pole positions and not to site directions of magnetization. This has been done and the results are given in Table 3. In Figure 6 a circle of radius 9°·1 equal to the standard deviation for the pole distribution is drawn and it can be seen that this contains the expected proportion

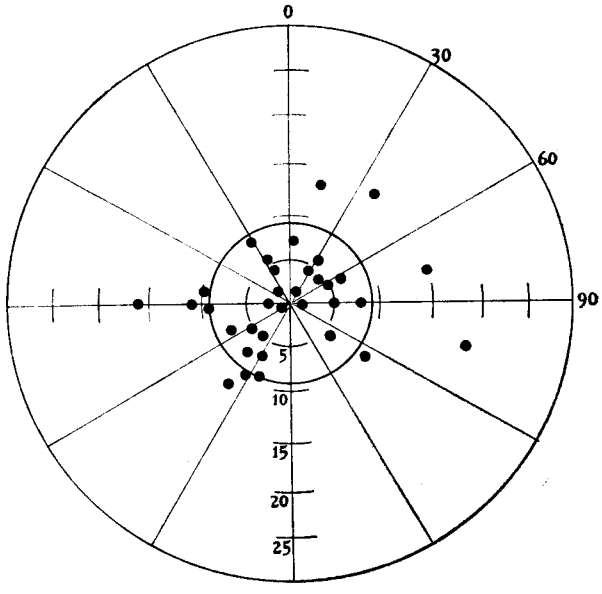


FIG. 6.—Site pole positions with the standard deviation circle (radius 9°·1). This circle contains 23 out of the 34 points.

(63 per cent) of the site poles. From equations (7) and (8) the values of the semi-axes of the standard deviation of the poles may be obtained. These are 9°·1 and 18°·2 respectively. This oval is drawn in Figure 3, and again it contains approximately 63 per cent of the observations. The pole calculated from the mean directions of magnetization of sites (168° E 37° N) does not differ appreciably from the mean pole position given in Table 3 since the mean of an oval distribution is still the sum of the direction cosines as for a circular distribution.

Table 3

The mean pole position of the Whin Sill

Latitude	Longitude	N	K	Standard deviation θ_0	Standard error θ_0	Error $P = 0{\cdot}05$ α
37°·3 N	168°·9 E	34	79·5	9°·1	1°·6	2°·8

There are several possible reasons why the distribution of directions in Figure 3 is oval.

(a) It was suggested by Graham & others (1957) that due to magnetostriction a n.r.m., acquired under non-hydrostatic load (e.g. in certain igneous rocks such as

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sills) changed on release of the stress due to erosional unloading or removal of the sample from the outcrop, and it is conceivable that such effects could have caused the oval distribution.

(b) The oval distribution may be due to a superimposed secondary component of the type discussed by Creer (1957). The most likely direction for such a secondary component is that of a dipole field along the present axis of rotation in which case the long axis of the oval distribution should be along the great circle connecting this direction to the mean direction of magnetization calculated above for the Whin Sill. This great circle is shown in Figure 6. Unfortunately it makes only a small angle with the palaeomagnetic meridian also marked in this figure so that it is not possible to say on geometrical grounds alone whether the data agree better with this interpretation than with that considered above. However the close agreement between many pole determinations from Europe (see Section 9) using rocks of different types and stress histories makes it unlikely that either (a) or (b) are important. Also it would be a remarkable coincidence if the effects of partial instability and stress deflections were of just the right magnitude and direction to produce a circular distribution of poles.

(c) Secular variation of the field could be responsible.

9. The Permian palaeomagnetic pole position relative to Europe

Data derived from palaeomagnetic studies of several formations in the Lower Permian of the British Isles, Norway, Germany and France are given in Table 4. The period of time represented probably spans much of the Lower Permian (Autunian and Saxonian = Rotliegende). The dating (which is by normal stratigraphical methods) of the intrusive igneous rocks is necessarily a little doubtful, especially with regard to some of the Norwegian and Ayrshire rocks. In the case of the Esterel igneous rocks and the Whin Sill, it is possible that some portion of the uppermost Carboniferous (Stephanian) is represented. The length of time covered by all formations is probably of the order of 10 million years. The calculated pole positions all lie in the Northern Pacific (Figure 7b). The site poles for the Whin Sill are given in Figure 7a for comparison and it is seen that they have a similar distribution.

It is of interest to investigate whether the apparent differences between these pole positions (Figure 7a) are significant or not. Such an investigation would give information about post-Permian relative movements between sampling areas or the extent of polar movements during the time represented. To do this it has first to be asked whether directions of magnetization have been sampled from a time interval large enough such that the means may be regarded as main dipole field directions. For those areas for which adequate data is available the standard deviation of the distribution of site mean directions of magnetization has been calculated in the same way as described for the Whin Sill in paragraph 4, and this is taken to represent the effects of secular variation during the interval of time when the formation in question acquired its magnetization. Consequently, it may be inferred that data for which the standard deviation of the distribution of site means is too small does not cover the full range of the ancient secular variation and therefore any geomagnetic pole calculated from it may be of the order of 10° away from the palaeogeographical pole. Reasons are given in paragraph 8 for suggesting that the length of the record revealed by the present work on the Whin Sill probably covers the full range of secular variation and the standard deviation of the

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Table 4

	Locality			Formation	Directions of n.r.m.		Standard deviation	Standard error	Number of samples	Number of sites	Position of Palaeomagnetic pole		Reference
					D	I	θ_0	θ_e	N_s	N_m			
British Isles	Ayrshire	55°·5 N	4°·5 W	Minor (Kylitic) intrusions	181°	+ 7°	16°·6	7°·4	11	6	32° N	174° E	Armstrong 1957
	Ayrshire	55°·5 N	4°·5 W	Mauchline sandstones	187°	− 6°	?	4°·6*	26	?	—	—	Du Bois 1957
	Ayrshire	55°·5 N	4°·5 W	Mauchline lavas	180°	− 4°	?	6°·7*	34	?	—	—	Du Bois 1957
	Northumberland and Durham	55° N	2° W	Whin Sill	188°	− 5°	13°·9	2°·4	102	34	37° N	169° E	This paper
	Devonshire	51° N	4° W	Exeter volcanic traps	189°	− 9°	20°·9	10°·4	33	5	43° N	164° E	Creer 1955, 1957
Norway	Oslo	60° N	10° E	Lavas & rhomboporphyry	202°	−33°	?	5°·5*	12	?	—	—	Rutten & others 1957
Germany	Pfalz	49° N	7°·5 E	Rotliegendes Ro	190°	+ 6°	16°·9	6°·0	13	9	38° N	175° E	Nairn (unpublished)
	Baden Baden	49° N	8° E	—(sandstones Ro-Rm)	184°	− 7°	12°·6	5°·5	10	6	45° N	182° E	Nairn (unpublished)
	Saar	49° N	10°·5 E	—R	181°	+ 4°	23°·2	11°·6	9	5	40° N	188° E	Nairn (unpublished)
	Freiburg	48° N	8° E	and lavas)—R	160°	+12°	18°·6	18°·6	4	2	—	—	Nairn (unpublished)
France	Vosges	48° N	6° E	Saxonian rhyolites	194°	− 6°	17°·7	7°·9	14	4	45° N	167° E	Nairn 1947
	Morvan	47° N	4° E	Saxonian sandstones					3	2			
	Esterel	43°·5 N	7° E	Pyromeride R_4	210°	−16°	?		5	?	—	—	Roche 1957
	Esterel	43°·5 N	7° E	Dolerites	175°	−13°	?		3	?	—	—	Roche 1957
	Esterel	43°·5 N	7° E	Rhyolite R_3	217°	+23°	?	4°·5	14	?	—	—	Rutten & others 1957

* Computed from 95 per cent error and given in the references.

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distribution of site mean directions of magnetization for this formation is 13°·9.* Creer (1955) has estimated the scatter to be expected due to present-day magnitude secular variation effects for different latitudes on the assumption that the distortions in the present field from that of an axial dipole field around any line of latitude are the same that any point on that line of latitude would experience over a time long with respect to the time constants of the core motions responsible for the secular variation. He obtained a value of $\theta_0 \simeq 20°$ for equatorial latitudes. This is comparable with many of the values* shown in Table 4, though some are rather lower. Whether this is because secular variation effects were less than at present during the time of magnetization of these rocks or because the time interval represented by them is short is unanswerable. Clearly at the present

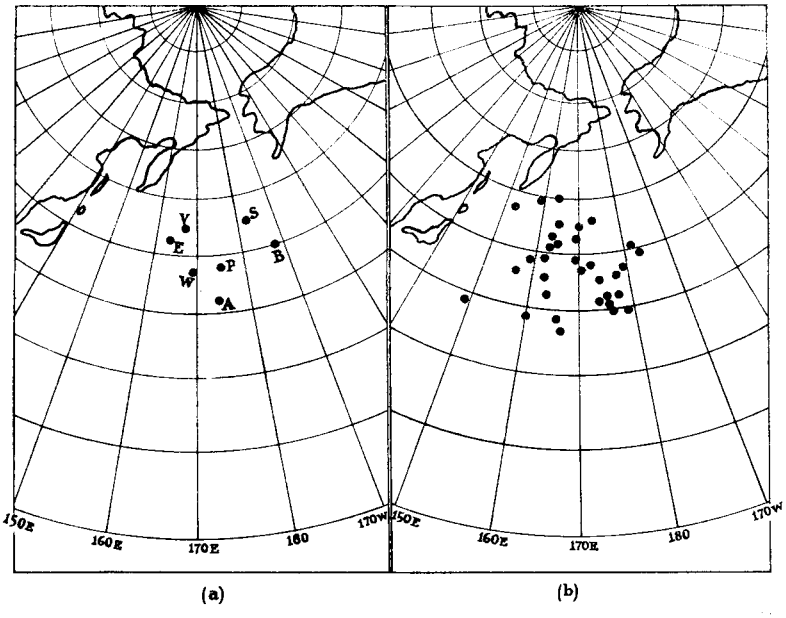


FIG. 7.—Maps showing the distribution of Lower Permian pole positions (a) and Whin Sill site pole positions (b).

stage it is not possible to say with certainty whether the differences between the pole determinations plotted in Figure 7a are real or are due to sampling inadequacies and although standard errors have been computed where possible, these should be treated with reserve at the moment, especially where the standard deviation is small or not given. The standard error used here is related to the standard deviation in the usual way and by the following formula $\theta_e = \theta_0/N_m^{\frac{1}{2}}$ where θ_e is the standard error, θ_0 the standard deviation of the distribution and N_m the number of sites which have been used in the computation of θ_0. The standard deviation is a measure of the width of the distribution and is one of its characteristic properties. With observations from a greater number of sites it can be determined with greater precision, but is not reduced numerically. (N_m must of course be large enough to represent the full range of secular variation as described above.) The standard error is a measure of the amount by which the

* In Table 4 θ_0 and θ_e have been calculated for the distribution of directions of magnetization and not from the distribution of poles given by these directions. This is because sufficient data was not available for recalculating all the localities quoted. For this reason it should be noted that the value given for θ_0 for Whin Sill differs from that given in Table 3.

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calculated mean direction may differ from the true direction and is reduced as the number of sites is increased. For a Fisherian distribution it is the 63 per cent error cone. 95 per cent errors (as often quoted in palaeomagnetic work) can be computed from the data for θ_0 given in Table 4 using the following formula adapted from Irving & Watson (1957):

$$\epsilon = 140/(KN_m)^{\frac{1}{2}} \tag{9}$$

and also equation (6).

10. Comparison with palaeoclimatic evidence

In Figure 8, Europe is drawn in the latitude and with the orientation suggested by the mean Permian result. On it are plotted the various climatic criteria; evaporites, desert sandstones, reef limestones, red beds and coal. These deposits

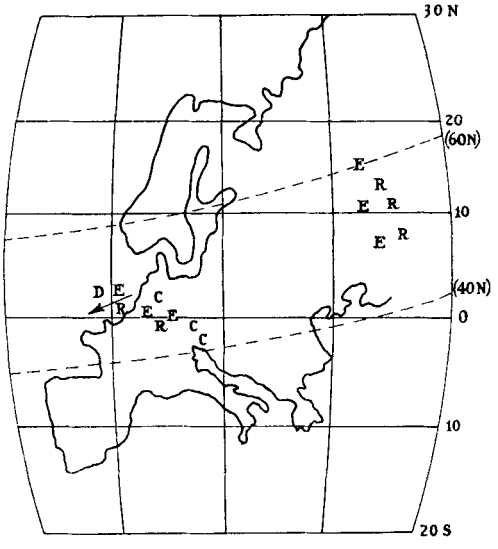


FIG. 8.—Map of Europe restored to the latitude and orientation suggested by Lower Permian palaeomagnetic results. The present 40° N and 60° N latitudes are retained for reference. Palaeoclimatic data are plotted using the following abbreviations: R reef corals, D desert sandstones, E evaporite and C coal deposits, arrow indicates the wind direction where known.

are usually thought to have been laid down in hot climates and their position close to the palaeomagnetic equator is consistent with the view that the geographic and mean magnetic poles coincide. (See general discussion in Irving 1956.) The wind direction inferred from the cross bedding in various dune sandstones (see Shotton 1956, Opdyke & Runcorn 1959) has a similar relation to the palaeomagnetic equator as the present day northern trade winds have to the present geographic equator. This is consistent with the proposed latitudes (see also Schove & others 1959) and suggests that the pole placed in the present day Pacific Ocean was the past geographic north pole assuming the present sense of rotation.

11. Conclusions

The Great Whin Sill has a stable, permanent magnetization with the site mean declination being 187°·8 with an inclination of −4°·9 confirming the earlier

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work of Manley (1949) and Hallimond & Butler (1949). The mean pole position in longitude 168°·9 E and latitude 37°·3 N is consistent with the recorded Lower Permian pole positions in western Europe.

The method of analysis used in this paper always gives circular polar errors, and has the advantage that correction is made for differences in latitude and longitude of sites. This is particularly useful when results from formations with great lateral spread are analysed, or when comparisons are made between the results from formations laid down during the same geological period in different parts of a continental land-mass.

The model proposed is broadly consistent with current models of the Earth's field, namely that of a steady axial dipole component constant for periods of 10^6 years perturbed by non-dipole components with periodicities of the order of 10^2–10^3 years. The strength of the dipole field at the poles is twice that in low latitudes whereas the non-dipole components are irregular and their magnitude does not appear to depend on latitude. Thus, in general, the effect of the latter on total direction of the field may be expected to be less in high latitudes than in low latitudes, as would be expected from (7) and (8). The oval distribution in the Whin Sill may therefore owe its form to the manner in which the non-dipole components offset the dipole field in low latitudes.

Oval distributions of n.r.m. may have been obscured in previous work because of random errors due to relative rotations between sites or inherent dispersion of magnetization or partial instability, and because sampling has not usually been undertaken in sufficient detail. Furthermore, where thick sequences have been studied, covering perhaps many millions of years, polar wandering effects (movements of the dipole axis (F)) may be superposed on the non-dipole variations. In the Whin Sill the time range covered is probably of the order of hundreds or thousands, not millions, of years so that polar wandering is unlikely to have occurred during the time it acquired its magnetization.

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